

Interface Definition Specification

Operator Tablet <-> Swarm — 35 Standard Messages

Interface Definition Specification (IDS)

문서번호 **SAN-IDS-CMD-001**

v1.5

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Classification	Interface Definition Specification
Preceding Docs	SAN-SDD-SWARM-001 v1.1
Message Format	Google Protobuf v3 + ROS 2 .msg
Transport	Wi-Fi 6 Mesh (OpenWrt) + Military LTE + LoRa
Message Count	39 (v1.0: 31 + 4 new in v1.1 + 2 new in v1.3 + 2 new in v1.5)

Revision History

Version	Changes
v1.0	Initial release. 31 standard messages (8 commands + 9 telemetry + 9 inter-robot + 5 infra).
v1.1	All 31 v1.0 messages preserved with the following changes: (a) Added VideoStreamRequest in §3 (Android pull-mode video request); (b) Modified VideoStreamHandle in §4 to specify GStreamer SRT/UDP; (c) Added SurveillanceSectorAssignment, PanTiltCommand, SLAMAggregatedMap in §5 (3 new); (d) Replaced SLAMDelta with SLAMLocalDelta in §5 (collection period 1Hz - > 5s (v1.3 shortened)). Total 35 messages.
v1.4	Deputy UGV message fields + new messages: (1) LeaderRoleAnnouncement message new (Section 5.15); (2) HubRoleAnnouncement message new (Section 5.16); (3) RobotStatus gains is_deputy_ugv, battery_percent fields; (4) deputy_chain default updated — [3, 2, ...] (Deputy 1st).
v1.5	DCN-2026-001 applied (2026-05-12): (1) §5.15 LeaderRoleAnnouncement body formally added (was Revision-History-only); (2) §5.16 HubRoleAnnouncement body formally added; (3) §5.14 LTERoleAnnouncement lte_backup_chain default extended [3, 5] → [3, 4, 5, 6, 7, 8] (all UGVs now ship with LTE); (4) §1.2 Total 35 → 39 (catalog correction: v1.3 §5.13/§5.14 + v1.5 §5.15/§5.16); (5) §8.3 Inter-Robot Messages catalog 12 → 16 rows; (6) Related Docs version pins updated from v1.1 to v1.5.

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1. Overview

1.1 Document Scope

This document defines every communication message of the SAN swarm system in a single specification. Scope includes:

1. Operator Android app -> Leader (command issuance)
2. Leader -> Operator Android app (state/telemetry)
3. Leader -> Follower (formation, predicted poses, surveillance sectors)
4. Follower -> Leader/Hub (state, SLAM aggregated map)
5. Hub UGV <-> wide-area C2 (military LTE backhaul)

1.2 Message Categories

Category	Count	Direction
§3 Operator Commands	9 (+1 new)	Tablet -> Leader
§4 Swarm Telemetry	9 (1 mod)	Swarm -> Tablet
§5 Inter-Robot	16 (+3 new in v1.1, +1 mod, +2 new in v1.3, +2 new in v1.5)	Leader <-> Follower <-> Hub
§6 Infrastructure/System	5	Bidirectional
Total	35	—

2. Message Architecture

2.1 Topic Naming Convention

<code>/ {robot_namespace} / {category} / {message_type}</code>	Examples:
<code>/leader/cmd/formation</code>	# operator -> leader
<code>/leader/cmd/surveillance_sectors</code>	# operator -> leader (sector config)
<code>/swarm/predict/follower_target</code>	# leader broadcast
<code>/swarm/surveillance/sector_assign</code>	# leader -> followers (10s period)
<code>/hub/slam/aggregated_map</code>	# Hub broadcast (5s (v1.3 shortened))
<code>/ {robot_id} /cmd/pantilt</code>	# leader or operator -> robot
<code>/tablet/cmd/video_request</code>	# tablet -> hub

2.2 Namespace Policy

Namespace	Purpose	Example Topic
/leader	Leader (Go2) exclusive	/leader/status, /leader/cmd
/hub	Hub UGV exclusive	/hub/slam, /hub/video_forward
/follower{N}	Each follower (N=1..7)	/follower1/status
/swarm	All-broadcast	/swarm/predict, /swarm/surveillance
/tablet	Operator app	/tablet/heartbeat

2.3 QoS 4-Tier

Priority	Reliability	Durability	History	Deadline	Example
P0 Critical	RELIABLE	TRANSIENT_LOCAL	KEEP_LAST(10)	100 ms	FormationCommand, EmergencyStop
P1 Tactical	BEST_EFFORT/REL	VOLATILE	KEEP_LAST(3)	200 ms	RobotStatus, FollowerTarget
P2 Operational	RELIABLE	TRANSIENT_LOCAL	KEEP_LAST(5)	1 s	SLAMLocalDelta, SLAMAggregatedMap
P3 Background	BEST_EFFORT	VOLATILE	KEEP_LAST(1)	5 s	LogEvent

3. Operator Command Messages (Tablet -> Leader/Hub)

3.1 FormationCommand

Attribute	Value
Direction	Tablet -> Leader
Topic	/leader/cmd/formation
QoS	P0, RELIABLE, KEEP_LAST(10)
Rate	Event

```
message FormationCommand {
  LINE=1; V_SHAPE=2; DIAMOND=3;
  VEE_INVERTED=7; FREE_SPREAD=8;
  float d = 2; // 1.5 <= d <= 20
  float theta_deg = 3;
  float transition_time = 5;
  bool timestamp_ms = 7;
  string requested_by = 8;
}

enum FormationType {
  COLUMN=0;
  ECHELON_LEFT=4; ECHELON_RIGHT=5; BOX=6;
}

FormationType type = 1;
float alpha_deg = 4;
float dev_mode = 6;
uint64 requested_by = 8;
```

3.2 MissionStateCommand

```
message MissionStateCommand {
  OFFENSIVE=2; DEFENSIVE=3;
  State target_state = 1;
  uint64 timestamp_ms = 3;
  string requested_by = 4;
}

enum State {
  IDLE=0; ASSEMBLY=1;
  PATROL=4; RTH=5; HOLD=6; EMERGENCY=7;
}

Pose2D target_position = 2;
uint64 requested_by = 4; // QoS: P0, RELIABLE
```

3.3 FireAuthorization

```
message FireAuthorization {
  uint32 assigned_robot = 2;
  THREE_BURST=1; FULL_AUTO=2; JAMMING=3;
  Pose2D target_pose = 4;
  uint64 expires_at_ms = 6;
  uint64 timestamp_ms = 7;
  string target_id = 1;
  enum FireMode {
    SINGLE_SHOT=0;
  }
  FireMode mode = 4;
  float lead_distance_m = 5;
  bytes operator_signature = 6;
  // QoS: P0, RELIABLE, KEEP_ALL
```

3.4 JammingCommand

```
message JammingCommand {
  BAND_5_8_GHZ=1; BAND_GNSS_L1=2; BAND_GNSS_L2=3; BAND_ALL=4;
  repeated Band bands = 1;
  float duration_s = 3;
  uint64 timestamp_ms = 5;
  enum Band {
    BAND_2_4_GHZ=0;
  }
  Pose2D target_direction = 2;
  float power_w = 4;
  // QoS: P0, RELIABLE
```

3.5 WaypointCommand

```
message WaypointCommand {
  float cruise_speed_mps = 2; // 0=auto, max 1.5
  bool allow_replan = 3;
  uint32 priority = 4;
  uint64 timestamp_ms = 5;
  repeated Pose2D waypoints = 1;
  // QoS: P1, RELIABLE
```

3.6 ManualOverrideCommand

```
message ManualOverrideCommand {      string  robot_id      = 1;      enum
Mode {          DISENGAGE_AUTO=0; DIRECT_DRIVE=1; HOLD_POSITION=2; FOLLOW_ME=3;
}          Mode      mode          = 2;          Velocity2D  manual_vel  = 3;
uint64  expires_at_ms  = 4;          uint64  timestamp_ms  = 5;      }      // QoS:
P0, RELIABLE
```

3.7 EmergencyStop

```
message EmergencyStop {      enum Severity {          SOFT_STOP=0;
HARD_STOP=1; DISARM=2; SHUTDOWN=3;      }          Severity  severity      = 1;
string  reason          = 2;          bytes    auth_signature  = 3;          uint64
timestamp_ms      = 4;      }      // QoS: P0, RELIABLE, KEEP_ALL, LoRa fallback ON
```

3.8 OperatorHeartbeat

```
message OperatorHeartbeat {      string  operator_id      = 1;          bytes
auth_token      = 2;          Pose2D   operator_pose      = 3;          uint32
app_version      = 4;          uint64   timestamp_ms      = 5;      }      // QoS: P1,
BEST_EFFORT, 1 Hz
```

3.9 VideoStreamRequest (v1.1 new)

Pull-mode request issued by the Android app. Prevents unnecessary bandwidth and power consumption.

Attribute	Value
Direction	Tablet -> Hub UGV (SBC #2)
Topic	/tablet/cmd/video_request
QoS	P1, RELIABLE
Rate	Event (operator taps View Video)

```
message VideoStreamRequest {      uint32  sequence          = 1;
uint32  target_robot_id      = 2;      string  protocol          = 3;      // "srt" |
"udp"          uint32  desired_port      = 4;          string  codec          = 5;
// "h265" | "h264"          string  quality          = 6;      // "hd" | "fhd" | "low" |
"thumbnail"          bool    encryption      = 7;          string  passphrase_hint
= 8;          string  action          = 9;      // "start" | "stop" | "change_quality"
uint64  timestamp_ms      = 10;      }
```

4. Swarm Telemetry Messages (Swarm -> Tablet)

4.1 RobotStatus

```
message RobotStatus {
  string robot_id = 1;
  enum Role {
    LEADER=0; HUB=1; FOLLOWER=2; } role = 2;
  Pose2D pose = 3;
  Velocity2D velocity = 4;
  float heading_rad = 5;
  enum Tier {
    T0_PREDICTIVE=0;
    T1_5_AUTO_REROUTE=1; T1_NORMAL=2;
    T2_CATCH_UP=3; T3_HARD_CATCH_UP=4;
    T4_BREADCRUMB=5; } Tier current_tier = 6;
  float battery_percent = 7;
  float battery_voltage = 8;
  float cpu_load_percent = 9;
  float mem_used_percent = 10;
  bool rtk_fixed = 11;
  bool comm_healthy = 12;
  bool lte_active = 13; // v1.3: this robot is the active LTE gateway
  bool is_lte_backup_designated = 14; // v1.3: designated as LTE backup follower
  uint32 errors_active = 13;
  uint64 timestamp_ms = 14; } //
QoS: P1, BEST_EFFORT, 1 Hz
```

4.2 FormationStatus

```
message FormationStatus {
  FormationType active_type = 1;
  float d = 2;
  float theta_deg = 3;
  MissionState mission_state = 4;
  uint32 total_robots = 5;
  uint32 healthy_robots = 6;
  repeated string unhealthy_ids = 7;
  float formation_error_m = 8;
  bool auto_column_active = 9;
  uint64 timestamp_ms = 10; } // QoS: P1, BEST_EFFORT, 2 Hz
```

4.3 DetectedObject

```
message DetectedObject {
  string object_id = 1;
  string detector_robot_id = 2;
  enum ObjectClass {
    PERSON=0; VEHICLE=1;
    DRONE_FPV=2; DRONE_MULTI=3;
    DRONE_FIXED=4; UNKNOWN=5; } obj_class = 3;
  float confidence = 4;
  Pose2D position = 5;
  Velocity2D velocity_estimate = 6;
  float range_m = 7;
  float bearing_deg = 8;
  bool is_threat = 9;
  uint32 threat_score = 10;
  bytes thumbnail_jpeg = 11;
  uint64 detected_at_ms = 12; } // QoS: P0, RELIABLE, event
```

4.4 ThreatAlert

```
message ThreatAlert {
  string alert_id = 1;
  enum Severity {
    ADVISORY=0; CAUTION=1; WARNING=2; CRITICAL=3; } severity = 2;
  enum ThreatType {
    DRONE_APPROACHING=0;
    ENEMY_PERSONNEL=1; ARTILLERY_DETECTED=2;
    AMBUSH_INDICATOR=3;
    COMM_JAMMING=4; GPS_JAMMING=5; } ThreatType threat_type = 3;
  float time_to_impact_s = 5;
  Pose2D string related_object_id = 4;
  float threat_position = 6;
  string message_text = 7;
  uint64 timestamp_ms = 8; } // QoS: P0, RELIABLE, KEEP_ALL
```

4.5 TierStatusChange

```
message TierStatusChange {
  string robot_id = 1;
  Tier from_tier = 2;
  Tier to_tier = 3;
  string
```

```

reason          = 4;          float    delta_m          = 5;          uint64
timestamp_ms    = 6;          }          // QoS: P0, RELIABLE, event

```

4.6 FireResult

```

message FireResult {          string    fire_id          = 1;          string
shooter_robot_id  = 2;          string    target_object_id  = 3;          uint32
rounds_fired      = 4;          bool      target_neutralized = 5;          float
hit_probability   = 6;          bytes     post_image_jpeg   = 7;          uint64
fired_at_ms      = 8;          }          // QoS: P0, RELIABLE

```

4.7 BatteryWarning

```

message BatteryWarning {          string    robot_id          = 1;          enum
Level { LOW=0; CRITICAL=1; SHUTDOWN=2; }          Level    level          = 2;
float    percent          = 3;          uint32    minutes_remaining = 4;          bool
should_rth          = 5;          uint64    timestamp_ms      = 6;          }          // QoS: P0,
RELIABLE

```

4.8 VideoStreamHandle (v1.1 modified — GStreamer specified)

Hub UGV returns video stream handle to APP. v1.0's simple RTSP URL is now extended to SRT/UDP URI + encryption key + actual bitrate.

```

message VideoStreamHandle {          uint32    sequence          = 1;
uint32    target_robot_id  = 2;          string    srt_uri          = 3;          // e.g.
"srt://hub_ip:8888?mode=listener"          string    codec          = 4;          // "h265"
| "h264"          string    quality          = 5;          // "hd" | "fhd" | "low" |
"thumbnail"          uint32    actual_bitrate_kbps = 6;          string    passphrase
= 7;          string    status          = 8;          // "streaming" | "stopped" | "error"
string    error_msg        = 9;          uint64    stream_start_ms      = 10;
uint64    timestamp_ms     = 11;          }          // QoS: P1, RELIABLE

```

4.9 SLAMUpdate

```

message SLAMUpdate {          Pose2D    origin          = 1;          float
resolution_m      = 2;          uint32    width_cells          = 3;          uint32
height_cells      = 4;          bytes     occupancy_delta      = 5;          uint32
contributing_robots = 6;          uint64    timestamp_ms          = 7;          }          // QoS: P2,
RELIABLE, 0.5 Hz (for tablet screen refresh)

```

5. Inter-Robot Messages

5.1 FollowerTargetMessage (1s prediction)

Attribute	Value
Direction	Leader -> Follower (broadcast)
Topic	/swarm/predict/follower_target
QoS	P0, RELIABLE, KEEP_LAST(3)
Rate	10 Hz (100 ms)
Size	~200 byte

```
message FollowerTargetMessage {
  uint32 follower_id = 1;
  Pose2D target_pose = 2;
  uint64 valid_at_ms = 3;
  repeated Pose2D leader_path_segment = 4;
  uint32 sequence_num = 5;
  float confidence = 6;
  Velocity2D target_velocity = 7;
  uint64 generated_at_ms = 8;
  message Pose2D {
    double x = 1;
    double y = 2;
    float heading = 3;
  }
  message Velocity2D {
    float vx = 1;
    float vy = 2;
    float omega = 3;
  }
```

5.2 BreadcrumbBroadcast

```
message BreadcrumbBroadcast {
  uint64 timestamp_us = 1;
  Pose2D position = 2;
  float speed_mps = 3;
  enum TerrainClass {
    OPEN=0; FOREST=1; URBAN=2; NARROW=3; CLIFF=4;
  }
  TerrainClass terrain = 4;
  uint8 formation_id = 5;
  uint32 sequence_num = 6;
} // QoS: P1, RELIABLE, KEEP_LAST(50), 0.5-2 Hz (1m moved or 0.5s)
```

5.3 LeaderElection (Modified Raft)

```
message LeaderElection {
  enum MessageType {
    REQUEST_VOTE=0;
    VOTE_GRANTED=1; VOTE_DENIED=2;
    LEADER_ANNOUNCE=3; HEARTBEAT=4;
  }
  MessageType type = 1;
  uint32 term = 2;
  string candidate_id = 3;
  float candidate_priority = 4;
  string voter_id = 5;
  uint64 timestamp_ms = 6;
} // QoS: P0, RELIABLE, KEEP_ALL, event (Hub UGV is priority 1)
```

5.4 EmergencyAlert

```
message EmergencyAlert {
  string source_robot_id = 1;
  enum AlertType {
    OBSTACLE_AHEAD=0; COLLISION_WARNING=1; CLIFF_DETECTED=2;
    FORMATION_BREAK=3; SENSOR_FAIL=4;
  }
  AlertType type = 2;
  Pose2D alert_position = 3;
  float severity = 4;
  string description = 5;
  uint64 timestamp_ms = 6;
} // QoS: P0, RELIABLE, event
```

5.5 SLAMLocalDelta (v1.1 modified — 1Hz -> 5s (v1.3 shortened) aggregated)

Replaces the v1.0 SLAMDelta (1 Hz). Sends an accumulated aggregated map every 5s (v1.3 shortened), reducing comm load by 1/100.

```
message SLAMLocalDelta {      uint32  sequence          = 1;
string  robot_id              = 2;      bytes  occupancy_grid_delta_png  = 3;
// PNG compressed          Pose2D  origin              = 4;      float
resolution_m                = 5;      uint64  coverage_start_ms          = 6;  //
aggregation start time      uint64  coverage_end_ms      = 7;  // aggregation
end time                    uint64  timestamp_ms         = 8;    }    // QoS: P2, RELIABLE
// Direction: all robots (Leader + 7 Follower)    // Receiver: Hub UGV SBC #1    //
Rate: 5s (v1.3 shortened) (default 30s, dynamic adjustment)
```

5.6 SLAMAggregatedMap (v1.1 new)

Broadcast of the merged global map by Hub UGV SBC #1. Updates the Nav2 cost map static layer on every robot.

```
message SLAMAggregatedMap {      uint32  sequence          = 1;      bytes
occupancy_grid_png  = 2;  // PNG compressed global map      Pose2D  origin
= 3;      float  resolution_m          = 4;      uint32  width_cells          = 5;
uint32  height_cells      = 6;      uint32  contributing_robots = 7;
uint64  timestamp_ms      = 8;    }    // QoS: P2, RELIABLE, TRANSIENT_LOCAL    //
Sender: Hub UGV (SBC #1)    // Receiver: all robots + operator tablet    // Rate: 5s
(v1.3 shortened) (dynamic)
```

5.7 GlobalMapBroadcast (v1.0 retained, operator screen)

```
message GlobalMapBroadcast {      Pose2D  map_origin          = 1;
float  resolution_m          = 2;      uint32  width              = 3;
uint32  height              = 4;      bytes  full_map              = 5;      bytes
delta_map                  = 6;      repeated string contributor_ids = 7;      uint64
timestamp_ms              = 8;    }    // QoS: P2, RELIABLE, TRANSIENT_LOCAL, 0.5 Hz
```

5.8 HeartBeat

```
message HeartBeat {      string  robot_id          = 1;      uint32
sequence_num          = 2;      uint64  uptime_ms          = 3;      float
cpu_load_percent      = 4;      bool    is_leader          = 5;      uint64
timestamp_ms          = 6;    }    // QoS: P1, BEST_EFFORT, 1 Hz
```

5.9 SlotAssignment

```
message SlotAssignment {      string  follower_id          = 1;      Pose2D
offset_in_body        = 2;      uint32  transition_id        = 3;      float
transition_time_s      = 4;      uint32  priority            = 5;      uint64
valid_from_ms          = 6;      uint64  timestamp_ms        = 7;    }    // QoS: P0,
RELIABLE, event (on formation change, Hungarian result)
```

5.10 SurveillanceSectorAssignment (v1.1 new)

Leader assigns surveillance sectors to each follower. Published every 10s (default) or immediately on events.

Attribute	Value
Direction	Leader -> all Followers + Hub UGV (broadcast)
Topic	/swarm/surveillance/sector_assign
QoS	P1, RELIABLE
Rate	10s (periodic) + event (formation change, threat)

```
message SurveillanceSectorAssignment {
    uint32 robot_id = 2;           // target
    float sector_start_deg = 3;    // -180 to 180 (heading-based)
    float sector_end_deg = 4;
    uint32 valid_period_sec = 5;   // default 10
    string priority = 6;           // "primary" | "threat_focus" | "overlap"
    string mode_hint = 7;          // "sweep" | "track" | "fixed"
    uint64 timestamp_ms = 8;
}
```

5.11 PanTiltCommand (v1.1 new)

Leader or operator controls a robot's pan-tilt. Followers may also self-publish to start sweep upon receiving sector assignment.

Attribute	Value
Direction	Leader or Tablet -> Robot
Topic	/robot_id/cmd/pantilt
QoS	P1, RELIABLE
Rate	Event or on sweep entry

```
message PanTiltCommand {
    uint32 robot_id = 2;           // heading-based
    float target_pan_deg = 3;
    float target_tilt_deg = 4;     // degrees/sec
    float speed_dps = 5;
    string mode = 6;               // "sweep" | "track" | "fixed" | "engage"
    float sweep_range_deg = 7;    // sweep width
    uint64 timestamp_ms = 8;
}
```

5.12 SwarmHealthSummary

```
message SwarmHealthSummary {
    uint32 total_robots = 1;
    uint32 online_robots = 2;
    float avg_battery_percent = 3;
    float min_battery_percent = 4;
    float avg_formation_err_m = 5;
    float comm_quality_percent = 6;
    uint32 robots_in_t4 = 7;
    uint32 robots_in_emergency = 8;
    uint64 timestamp_ms = 9;
}
// QoS: P1, BEST_EFFORT, 1 Hz, Hub -> Tablet
```

5.13 CostMapUpdate (v1.3 new)

Each UGV publishes its computed Local Cost Map. Hub UGV SBC #1 collects them for cross-validation.

Attribute	Value
Direction	Each UGV -> broadcast
Topic	/ {robot_id} /local/cost_map
QoS	P2, BEST_EFFORT, KEEP_LAST(1)
Rate	1 Hz (KPP: cost_map_latency_p99 <= 5s)

5.14 LTERoleAnnouncement (v1.3 new)

LTE gateway promotion/demotion broadcast. When Hub UGV (S2) LTE fails, Follower S3 (1st priority) auto-promotes.

Attribute	Value
Direction	LTE backup promoting/demoting robot -> broadcast
Topic	/swarm/lte/role_announce
QoS	P0, RELIABLE, KEEP_ALL
Rate	Event (on promotion/demotion/re-affirm)

5.15 LeaderRoleAnnouncement (★ v1.5 new — body formally added per DCN-2026-001 D-005)

Driver of the 4-tier Leader succession chain defined in SAN-SDD-SWARM-001 v1.5 §5.6. Each candidate self-declares with a CANDIDATE announcement after a priority-tier grace period (200 ms × priority); the highest-priority candidate wins by emitting a PROMOTED. Split-brain prevention via monotonically increasing leader_term.

Attribute	Value
Direction	Candidate / new leader → broadcast
Topic	/swarm/leader/role_announce
QoS	P0, RELIABLE, TRANSIENT_LOCAL, KEEP_LAST(10)
Rate	Event (on CANDIDATE / PROMOTED / DEMOTED transition)
Implementing package	san_role_management
Tested in	SAN-TST-INT-001 v1.5 S18-1, S18-2, S18-4

// SAN v1.5 IDS §5.15 – Leader 4-tier succession broadcast

```

message LeaderRoleAnnouncement {
  enum Role {
    LEADER_NORMAL      = 0;    // current leader operating normally, or non-leader
robot
    LEADER_CANDIDATE = 1;    // self-declared candidate, grace-period in progress
    LEADER_PROMOTED  = 2;    // succession complete, this robot is current leader
    LEADER_DEMOTED   = 3;    // original leader recovered, this robot demoted
  }
  enum SuccessionPriority {
    PRIORITY_UNSPECIFIED = 0;
    PRIORITY_DEPUTY      = 1;    // 1st: Deputy UGV (S3)
    PRIORITY_HUB         = 2;    // 2nd: Hub UGV (S2)
    PRIORITY_BATTERY_MAX = 3;    // 3rd: max-battery follower
    PRIORITY_LIMP_MODE   = 4;    // 4th: no eligible candidate → Limp Mode
  }
  Header          header          = 1;
  uint32          sequence         = 2;
  uint32          robot_id         = 3;
  uint32          leader_term      = 4;    // monotonic, split-brain prevention
  Role            role             = 5;
  SuccessionPriority succession_priority = 6;
  float           battery_percent  = 7;    // 0.0 ~ 100.0 (priority 3
comparison)
  string          reason           = 8;    // "leader_heartbeat_timeout" |
"leader_recovered" | "battery_lowest"
  uint64          timestamp_ms     = 9;
}

```

5.15.1 Split-brain Prevention Rule

Receivers compare incoming `leader_term` with their cached value: (a) incoming < cached → discard as stale, log WARN; (b) incoming > cached → adopt new term and respect role transition; (c) incoming = cached, different `robot_id` → lowest `robot_id` wins (tie-breaker).

5.15.2 Grace Period Timing

Leader heartbeat timeout = 1400 ms (200 ms × 7 missed samples). On timeout, candidates wait `priority × 200 ms` before self-promoting: DEPUTY (S3) 200 ms grace → promotes at $t+1.6$ s after Leader failure; HUB (S2) 400 ms grace → promotes at $t+1.8$ s only if no DEPUTY announcement seen; BATTERY_MAX 600 ms grace → promotes at $t+2.0$ s only if no DEPUTY/HUB announcement.

5.16 HubRoleAnnouncement (★ v1.5 new — body formally added per DCN-2026-001 D-005)

Hub UGV (S2) heartbeat 5-second timeout triggers Deputy UGV (S3) to broadcast HUB_PROMOTED while assuming LTE gateway + SLAM aggregation + GStreamer SRT relay duties (full takeover). On Hub recovery, the Deputy emits HUB_DEMOTED. If

both Hub and Deputy are down, the LimpModeManager (see SAN-SDD-SWARM-001 v1.5 §5.7) detects the gap and broadcasts LIMP_MODE_ACTIVE on a separate channel — no HUB_PROMOTED is emitted in that case.

Attribute	Value
Direction	Promoting / demoting Hub-role robot → broadcast
Topic	/swarm/hub/role_announce
QoS	P0, RELIABLE, TRANSIENT_LOCAL, KEEP_LAST(10)
Rate	Event (on FAILING / PROMOTED / DEMOTED transition)
Implementing package	san_role_management
Tested in	SAN-TST-INT-001 v1.5 S18-3, S18-5, S18-6

// SAN v1.5 IDS §5.16 – Hub-Deputy 이중화 broadcast

```
message HubRoleAnnouncement {
    enum Role {
        HUB_NORMAL    = 0;    // current Hub operating normally, or non-Hub robot
        HUB_FAILING    = 1;    // Hub heartbeat delay detected (Deputy succession pending)
        HUB_PROMOTED   = 2;    // Deputy assumed Hub role (LTE + SLAM + Video all
transferred)
        HUB_DEMOTED    = 3;    // original Hub recovered, Deputy demoted
    }
    Header header                = 1;
    uint32 sequence              = 2;
    uint32 robot_id              = 3;
    uint32 hub_term              = 4;    // monotonic, split-brain prevention
    Role role                    = 5;
    bool  lte_active              = 6;    // LTE gateway active
    bool  slam_aggregation_active = 7;    // SLAM aggregation active
    bool  video_relay_active      = 8;    // GStreamer SRT relay active
    string reason                 = 9;    // "hub_heartbeat_timeout" | "hub_recovered"
| "init"
    uint64 timestamp_ms          = 10;
}
```

5.16.1 Full Takeover Semantics

"HUB_PROMOTED" implies all three capability flags are simultaneously true. Partial takeovers (only LTE, or only SLAM) are NOT valid Hub roles — they would be advertised via LTERoleAnnouncement (§5.14) only.

5.16.2 Recovery Sequence

When original Hub (S2) heartbeat resumes: (1) S2 broadcasts HubRoleAnnouncement(HUB_NORMAL, hub_term=M+1) with intent to reclaim; (2) Deputy (S3) compares S2 batteries / SBC health against own; if S2 nominal, S3 broadcasts HUB_DEMOTED with hub_term=M+1 and ceases LTE/SLAM/Video relay; (3) S2 resumes Hub duties; S3 returns to standby. Verified in TST S18-6 (≤ 5 s exit).

6. Infrastructure/System Messages

6.1 TopicCatalog

```
message TopicCatalog {      string  robot_id      = 1;      repeated
TopicEndpoint publishers  = 2;      repeated TopicEndpoint subscribers = 3;
uint64 generated_at_ms    = 4;      }      message TopicEndpoint {      string
topic_name      = 1;      string  message_type  = 2;      string  qos_profile
= 3;      }      // QoS: P1, RELIABLE, TRANSIENT_LOCAL, event (startup/change)
```

6.2 VersionNegotiation

```
message VersionNegotiation {      string      peer_id      = 1;
uint32      schema_major      = 2;      uint32      schema_minor      = 3;
uint32      sw_version      = 4;      repeated uint32 supported_majors = 5;
bool      is_compatible      = 6;      string      error_message      = 7;
uint64      timestamp_ms      = 8;      }      // QoS: P0, RELIABLE
```

6.3 SecurityHandshake

```
message SecurityHandshake {      string  peer_id      = 1;
bytes  certificate_chain  = 2;      // X.509 v3      bytes  nonce      =
3;      bytes  signature      = 4;      // RSA 4096      string  cipher_suite
= 5;      // "AES-256-GCM"      uint64  timestamp_ms      = 6;      }      // QoS: P0,
RELIABLE
```

6.4 LogEvent

```
message LogEvent {      string  source_id      = 1;      enum Severity {
DEBUG=0; INFO=1; WARN=2; ERROR=3; FATAL=4;      }      Severity severity
= 2;      string  component      = 3;      string  message      = 4;
string  correlation_id  = 5;      uint64  timestamp_ms      = 6;      }      // QoS:
P3, BEST_EFFORT
```

6.5 TimeSync

```
message TimeSync {      uint64  master_time_us      = 1;      int64
offset_us      = 2;      float  drift_ppm      = 3;      uint32
sync_round      = 4;      }      // QoS: P0, BEST_EFFORT, 1 Hz, Hub master -> all
robots      // PTP IEEE-1588 grandmaster on Hub UGV SBC #1
```

7. Security (DDS-Security)

Function	Standard	Adopted Value
Authentication	OMG DDS-Security AuthN	PKI X.509 v3, RSA 4096
Encryption	OMG DDS-Security Crypto	AES-256-GCM
Access Control	OMG DDS-Security AccessCtrl	Permissions Document
Logging	OMG DDS-Security Logging	Centralized on Hub UGV
Key Management	Internal policy	Hub UGV TPM, per-mission rotation

7.1 Video Stream Security

- SRT built-in AES-128 encryption (per-mission passphrase)
- Hub-Operator segment: IPsec or OpenVPN tunnel (additional)
- Robot-Hub LAN: WPA3-SAE + DDS Security

7.2 Emergency Key Revocation

On suspected robot loss/capture, the operator app's "Compromise" button immediately revokes that robot's certificate. The CRL is broadcast to the entire swarm immediately.

8. Appendix — Message Catalog (35 total)

8.1 Command Messages (9)

#	Name	Topic	QoS	Rate	v1.1
3 · 1	FormationCommand	/leader/cmd/formation	P0	evt	
3 · 2	MissionStateCommand	/leader/cmd/mission_state	P0	evt	
3 · 3	FireAuthorization	/leader/cmd/fire_auth	P0	evt	
3 · 4	JammingCommand	/leader/cmd/jamming	P0	evt	
3 · 5	WaypointCommand	/leader/cmd/waypoint	P1	evt	
3 · 6	ManualOverrideCommand	/robot/cmd/manual_override	P0	evt	
3 · 7	EmergencyStop	/swarm/emergency_stop	P0	evt	
3 · 8	OperatorHeartbeat	/leader/heartbeat/operator	P1	1 Hz	
3 · 9	VideoStreamRequest	/tablet/cmd/video_request	P1	evt	* new

8.2 Telemetry Messages (9)

#	Name	Topic	QoS	Rate	v1.1
4 · 1	RobotStatus	/robot/status/robot	P1	1 Hz	
4	FormationStatus	/leader/status/formation	P1	2 Hz	

#	Name	Topic	QoS	Rate	v1.1
2					
4 3	DetectedObject	/robot/detected/object	P0	evt	
4 4	ThreatAlert	/leader/alert/threat	P0	evt	
4 5	TierStatusChange	/robot/status/tier_change	P0	evt	
4 6	FireResult	/robot/status/fire_result	P0	evt	
4 7	BatteryWarning	/robot/warn/battery	P0	evt	
4 8	VideoStreamHandle	/leader/status/video_handle	P1	evt	* mod
4 9	SLAMUpdate	/hub/slam/global_map_delta	P2	0.5 Hz	

8.3 Inter-Robot Messages (16)

#	Name	Topic	QoS	Rate	v1.1
5 1	FollowerTargetMessage	/swarm/predict/follower_target	P0	10 Hz	
5 2	BreadcrumbBroadcast	/swarm/breadcrumb	P1	0.5-2 Hz	
5 3	LeaderElection	/swarm/leader_election	P0	evt	
5 4	EmergencyAlert	/swarm/alert/emergency	P0	evt	

#	Name	Topic	QoS	Rate	v1.1
4					
5 . 5	SLAMLocalDelta	/robot/slam/local_delta	P2	5s (v1.3 shortened)	* mod
5 . 6	SLAMAggregatedMap	/hub/slam/aggregated_map	P2	5s (v1.3 shortened)	* new
5 . 7	GlobalMapBroadcast	/swarm/slam/global_map	P2	0.5 Hz	
5 . 8	HeartBeat	/robot/heartbeat	P1	1 Hz	
5 . 9	SlotAssignment	/follower/cmd/slot_assignment	P0	evt	
5 . 10	SurveillanceSectorAssignment	/swarm/surveillance/sector_assign	P1	10 s	* new
5 . 11 1	PanTiltCommand	/robot/cmd/pantilt	P1	evt	* new
5 . 12	SwarmHealthSummary	/hub/health/swarm_summary	P1	1 Hz	

8.4 Infrastructure/System Messages (5)

#	Name	Topic	QoS	Rate	v1.1
6	TopicCatalog	/robot/system/	P1	evt	

#	Name	Topic	QoS	Rate	v1.1
1		topic_catalog			
6 2	VersionNegotiation	/swarm/system/version_neg	P0	evt	
6 3	SecurityHandshake	/swarm/system/ security_handshake	P0	evt	
6 4	LogEvent	/robot}/system/log	P3	evt	
6 5	TimeSync	/swarm/system/time_sync	P0	1 Hz	

8.5 v1.0 -> v1.1 Change Summary

Change	Messages	Rationale
New (+4)	VideoStreamRequest, SurveillanceSectorAssignment, PanTiltCommand, SLAMAggregatedMap	v1.1 four design changes
Modified (2)	VideoStreamHandle (GStreamer SRT specified), SLAMLocalDelta (replaces SLAMDelta, 5s (v1.3 shortened))	Protocol specification + period adjustment
Retained (29)	All other v1.0 messages	—
Total	35 messages	—